

2024-2025 Spring
MAT124 Midterm
09/04/2025

1. Consider the curve with the polar equation $r = 2 \cos(3\theta)$.
 - a. Find symmetry properties for the given curve and sketch the curve.
 - b. Find the area of the region enclosed by the curve.
2. Identify (describe and sketch) the following curves with polar equations using Cartesian coordinates.
 - a. $r = 2 \sin \theta + 2 \cos \theta$
 - b. $r = \tan \theta \sec \theta$
3. a. The following vectors are given.

$$\mathbf{u} = 6\mathbf{i} - 3\mathbf{j} + 2\mathbf{k} \quad \text{and} \quad \mathbf{v} = (4p + 1)\mathbf{i} + (p - 2)\mathbf{j} + \mathbf{k}$$

p is a scalar constant. Find the value of p if

- i. $\mathbf{u}$ and $\mathbf{v}$ are perpendicular vectors.
 - ii. $\mathbf{u}$ and $\mathbf{v}$ are parallel vectors.
 - b. The points $A(5, 1, 3)$, $B(3, 1, 5)$, $C(5, 3, 5)$ are given. Show that the triangle $\triangle ABC$ is equilateral and find its area.
 - c. Find the intersection of the line L and the plane R .
4. a. Find an equation of the plane through the point $P(1, 2, 3)$ parallel to the plane $R: x + y + z = 1$.
b. Find the parametric equations of a line L through the point $P(1, 2, 3)$ and perpendicular to the plane R .
5. Evaluate the limits, if they exist, and explain your answer.

a. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{x^6 + y^2}$

b. $\lim_{(x,y) \rightarrow (0,0)} x^4 \sin\left(\frac{1}{x^2 + |y|}\right)$

6. a. Find a vector function that represents the curve of intersection of two cylinders $x^2 + y^2 = 1$ and $z = 4x^2$.
b. Find the unit tangent vector to the curve in part (a) at $t = \frac{\pi}{2}$.
c. Find a formula for the length of the curve in part (a). Do not evaluate the length.
7. Use traces to sketch and identify the surface given by the equation $z = -x^2 - y^2 + 2$.

1. a. Determine whether the graph is symmetric about the x -axis.

$$(r, \theta) \rightarrow r = 2 \cos(3\theta), \quad (r, -\theta) \rightarrow r = 2 \cos(-3\theta) = 2 \cos(3\theta)$$

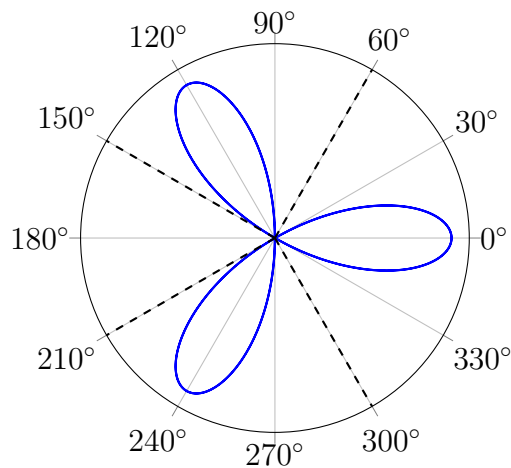
(r, θ) is on the graph, and the graph is symmetric about the x -axis. Determine whether the graph is symmetric about the origin.

$$(r, \theta) \rightarrow r = 2 \cos(3\theta), \quad (r, \theta + \pi) \rightarrow r = 2 \cos(-3(\theta + \pi)) = -2 \cos(3\theta)$$

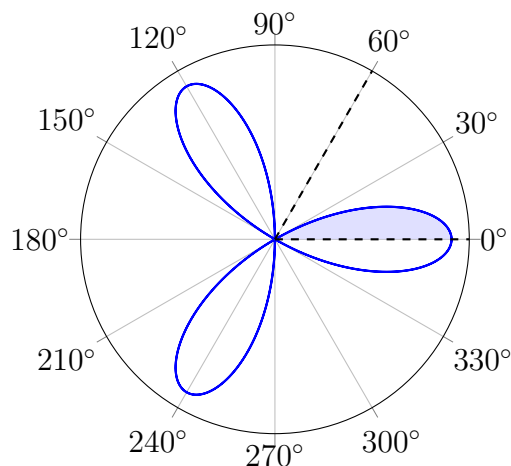
$(r, \theta + \pi)$ is not on the graph, and the graph is not symmetric about the origin. Determine whether the graph is symmetric about the y -axis.

$$(r, \theta) \rightarrow r = 2 \cos(3\theta), \quad (r, \theta - \pi) \rightarrow r = 2 \cos(-3(\theta - \pi)) = -2 \cos(3\theta)$$

$(r, \theta - \pi)$ is not on the graph, and the graph is not symmetric about the y -axis.



- b. It is sufficient to calculate the area of the upper half of the leaf right to the y -axis and multiply the result by six.



$$\begin{aligned} \frac{1}{2} \int_0^{\pi/6} (2 \cos(3\theta))^2 d\theta &= \frac{1}{2} \int_0^{\pi/6} 4 \cos^2(3\theta) d\theta = 2 \int_0^{\pi/6} \frac{1 + \cos(6\theta)}{2} d\theta \\ &= \theta + \frac{\sin(6\theta)}{6} \Big|_0^{\pi/6} = \frac{\pi}{6} \end{aligned}$$

The area is then

$$\text{Area} = 6 \cdot \frac{\pi}{6} = \boxed{\pi}$$

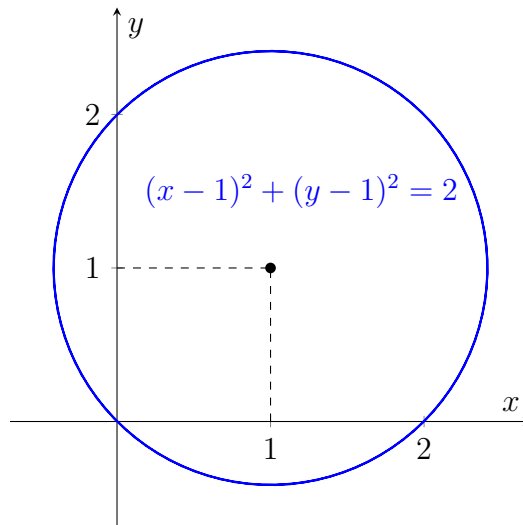
2. a. Multiply each side by r .

$$r = 2 \sin \theta + 2 \cos \theta \implies r^2 = 2r \sin \theta + 2r \cos \theta$$

Using the equations $x = r \cos \theta$ and $y = r \sin \theta$, we get

$$\begin{aligned} x^2 + y^2 &= 2x + 2y \implies x^2 - 2x + y^2 + 2y = 0 \implies x^2 - 2x + 1 + y^2 - 2y + 1 = 2 \\ &\implies (x - 1)^2 + (y - 1)^2 = (\sqrt{2})^2 \end{aligned}$$

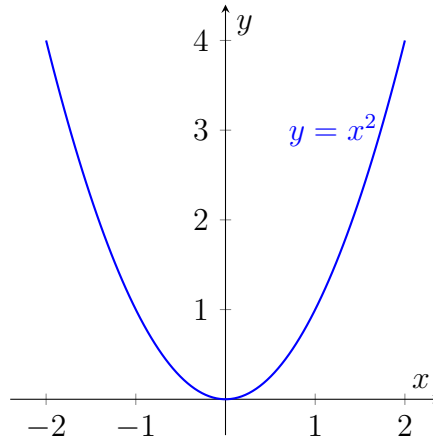
This is a circle with radius $\sqrt{2}$ centered at $(1, 1)$.



b. Rearrange the equation.

$$r = \tan \theta \sec \theta = \frac{\sin \theta}{\cos^2 \theta} \implies r \cos^2 \theta = \sin \theta \implies r^2 \cos^2 \theta = r \sin \theta$$

Using the equations $x = r \cos \theta$ and $y = r \sin \theta$, we get the parabola $x^2 = y$, where its branches open upward along the y -axis with the vertex $(0, 0)$.



3. a. i. If $\mathbf{u}$ and $\mathbf{v}$ are perpendicular, the dot product of these vectors is equal to zero.

$$\mathbf{u} \cdot \mathbf{v} = \langle 6, -3, 2 \rangle \cdot \langle 4p + 1, p - 2, 1 \rangle = 6(4p + 1) - 3(p - 2) + 2 \cdot 1 = 21p + 14 = 0$$

$$\implies \boxed{p = -\frac{2}{3}}$$

- ii. If $\mathbf{u}$ and $\mathbf{v}$ are parallel, the cross product of these vectors is equal to the zero vector.

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 6 & -3 & 2 \\ 4p + 1 & p - 2 & 1 \end{vmatrix} \\ &= \mathbf{i} \begin{vmatrix} -3 & 2 \\ p - 2 & 1 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 6 & 2 \\ 4p + 1 & 1 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 6 & -3 \\ 4p + 1 & p - 2 \end{vmatrix} \\ &= [1 \cdot (-3) - 2(p - 2)] \mathbf{i} - (6 \cdot 1 - 2(4p + 1)) \mathbf{j} + \\ &\quad + [6(p - 2) - (-3)(4p + 1)] \mathbf{k} \\ &= (1 - 2p) \mathbf{i} - (4 - 8p) \mathbf{j} + (-9 + 18p) \mathbf{k} = \mathbf{0} \implies \boxed{p = \frac{1}{2}} \end{aligned}$$

- b. The sides of an equilateral triangle have the same length.

$$\overrightarrow{AB} = \langle 3 - 5, 1 - 1, 5 - 3 \rangle = \langle -2, 0, 2 \rangle \rightarrow |\overrightarrow{AB}| = \sqrt{(-2)^2 + 0^2 + 2^2} = \sqrt{8}$$

$$\overrightarrow{AC} = \langle 5 - 5, 3 - 1, 5 - 3 \rangle = \langle 0, 2, 2 \rangle \rightarrow |\overrightarrow{AC}| = \sqrt{0^2 + 2^2 + 2^2} = \sqrt{8}$$

$$\overrightarrow{BC} = \langle 5 - 3, 3 - 1, 5 - 5 \rangle = \langle 2, 2, 0 \rangle \rightarrow |\overrightarrow{BC}| = \sqrt{2^2 + 2^2 + 0^2} = \sqrt{8}$$

Half of the magnitude of the cross product of two vectors gives us the area of the

triangle.

$$\frac{1}{2} \left| \overrightarrow{AB} \times \overrightarrow{AC} \right| = \frac{1}{2} \left| \overrightarrow{AB} \right| \left| \overrightarrow{AC} \right| \sin \frac{\pi}{3} = \frac{1}{2} \cdot \sqrt{8} \cdot \sqrt{8} \cdot \frac{\sqrt{3}}{2} = \boxed{2\sqrt{3}}$$

4. a. The planes have the same normal $\mathbf{n} = \langle 1, 1, 1 \rangle$. Using the definition $\mathbf{n} \cdot \overrightarrow{PP_0} = 0$, we obtain

$$1(x - 1) + 1(y - 2) + 1(z - 3) = 0 \implies \boxed{x + y + z = 6}$$

- b. The direction vector of the line is the normal vector of the plane. Therefore, the parametric equations for the line L is as follows.

$$\left. \begin{array}{l} x = 1 + t \\ y = 2 + t \\ z = 3 + t \end{array} \right\} \quad t \in \mathbb{R}$$

- c. Substitute the equation of L in the equation of the plane R .

$$x + y + z = 1 \implies (1 + t) + (2 + t) + (3 + t) = 1 \implies 3t + 6 = 1 \implies t = -\frac{5}{3}$$

$$t = -\frac{5}{3} \implies x = 1 - \frac{5}{3} = -\frac{2}{3}, \quad y = 2 - \frac{5}{3} = \frac{1}{3}, \quad z = 3 - \frac{5}{3} = \frac{4}{3}$$

Therefore, the point of intersection is $\boxed{\left(-\frac{2}{3}, \frac{1}{3}, \frac{4}{3}\right)}$.

5. a. Apply the Two-Path Test.

$$y = x \implies \lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{x^6 + y^2} = \lim_{x \rightarrow 0} \frac{x^4}{x^6 + x^2} = \lim_{x \rightarrow 0} \frac{x^2}{x^4 + 1} = \frac{0}{1} = 0$$

$$y = x^3 \implies \lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{x^6 + y^2} = \lim_{x \rightarrow 0} \frac{x^6}{2x^6} = \lim_{x \rightarrow 0} \frac{1}{2} = \frac{1}{2}$$

Since $0 \neq \frac{1}{2}$, by the Two-Path Test, the limit does not exist.

- b. We have the inequality $|\sin \theta| \leq 1$ for all values of θ . Therefore,

$$-1 \leq \sin \left(\frac{1}{x^2 + |y|} \right) \leq 1 \quad [\text{except for } x = 0, y = 0]$$

$$-x^4 \leq x^4 \sin \left(\frac{1}{x^2 + |y|} \right) \leq x^4$$

$$\lim_{(x,y) \rightarrow (0,0)} -x^4 = \lim_{(x,y) \rightarrow (0,0)} x^4 = 0 \implies \lim_{(x,y) \rightarrow (0,0)} x^4 \sin \left(\frac{1}{x^2 + |y|} \right) = \boxed{0}$$

By the squeeze theorem, the limit is equal to zero.

6. a. Parametrize the curve using $0 \leq t \leq 2\pi$.

$$x = \cos t, y = \sin t, z = 4 \cos^2 t, \quad 0 \leq t \leq 2\pi$$

$$\mathbf{r}(t) = \langle \cos t, \sin t, 4 \cos^2 t \rangle \quad 0 \leq t \leq 2\pi$$

- b. The tangent vector can be obtained by taking the first derivative of the vector function.

$$\mathbf{T}(t) = \mathbf{r}'(t) = \langle -\sin t, \cos t, -8 \cos t \sin t \rangle$$

The unit tangent vector is

$$\frac{\mathbf{T}(t)}{|\mathbf{T}(t)|} = \frac{\langle -\sin t, \cos t, -8 \cos t \sin t \rangle}{\sqrt{(-\sin t)^2 + (\cos t)^2 + (-8 \cos t \sin t)^2}} = \frac{\langle -\sin t, \cos t, -8 \cos t \sin t \rangle}{\sqrt{1 + 64 \cos^2 t \sin^2 t}}$$

At $t = \frac{\pi}{2}$,

$$\frac{\mathbf{T}(t = \pi/2)}{|\mathbf{T}(t = \pi/2)|} = \frac{\langle -\sin \frac{\pi}{2}, \cos \frac{\pi}{2}, -8 \cos \frac{\pi}{2} \sin \frac{\pi}{2} \rangle}{\sqrt{1 + 64 \cos^2 \frac{\pi}{2} \sin^2 \frac{\pi}{2}}} = \langle -1, 0, 0 \rangle$$

- c. The length of the parametrized curve $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ for $a \leq t \leq b$ can be evaluated using the integral

$$L = \int_a^b \left| \frac{d\mathbf{r}}{dt} \right| dt$$

The length of the curve is then

$$L = \int_0^{2\pi} \left| \frac{d\mathbf{r}}{dt} \right| dt = \int_0^{2\pi} \sqrt{1 + 64 \cos^2 t \sin^2 t} dt = \int_0^{2\pi} \sqrt{1 + 16 \sin^2 2t} dt$$

7. This is a circular paraboloid with the vertex $(0, 0, 2)$ opening downward along the z -axis.

